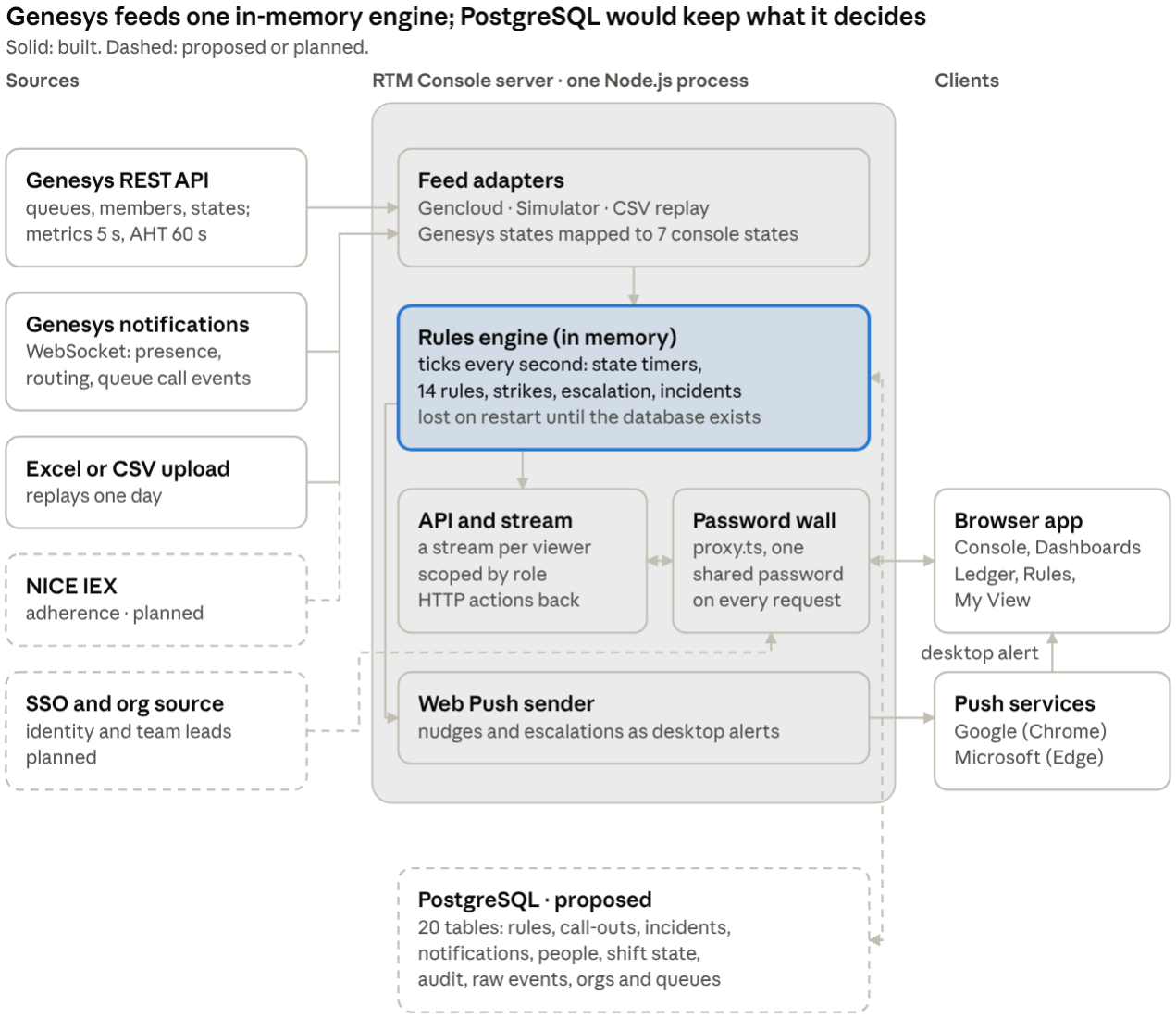
RTM Console Architecture and Data Sources

Oct 12, 2026 · @Adrian

The console is one Node.js server that pulls live agent and queue data from Genesys Cloud, runs the rules engine in memory, and streams each viewer's span to the browser. PostgreSQL is the proposed store for everything that must survive a restart; today nothing is saved.

Architecture

One server process holds a single live floor that every viewer shares. Data flows in from Genesys, through the rules engine, out to browsers over a server-sent event stream; user actions come back as ordinary HTTP requests. The dashed parts are proposed or planned.



architecture · built and proposed

The browser and server talk through the password wall in both directions: the stream comes out, actions go in. The engine's link to PostgreSQL is the part to build next.

Part	Where in the code	What it does	Status
Genesys Cloud	External	Agent presence and routing, queue membership, queue metrics	Live
Gencloud feed adapter	`lib/feed/GencloudFeed.ts`, `lib/feed/gencloud/`	Loads the watched queues and the roster (active accounts with a login in the last 30 days), reads every agent's current state in bulk, keeps WebSockets open for presence, routing and queue call events, polls queue metrics every 5 s and agent handle time every 60 s	Built
Simulator and replay	`lib/feed/SimFeed.ts`, `lib/feed/CsvReplayFeed.ts`	A stand-in floor with real names; replays one day from the Excel template or a Genesys CSV	Built
Rules engine	`lib/engine/`	Ticks every second: state timers, 14 rules, strikes, the escalation ladder, incidents	Built, in memory
Runtime	`lib/server/runtime.ts`	One live floor for everyone, a private runtime per replay, the Genesys/simulation switch	Built

Part	Where in the code	What it does	Status
Token store	`lib/server/tokenStore.ts`, `/admin/token`	Saves a pasted Genesys token on the server and reconnects the feed without a restart; guarded by `RTM_ADMIN_SECRET`	Built
Floor clock	`lib/floorTime.ts`	Shift clock and the start of "today" on US Eastern time (`RTM_TIME_ZONE`), not the server's Manila time	Built
Roster cache	`lib/server/rosterCache.ts`	Keeps the last Genesys roster's names in `.rtm/roster.json` so the simulator can use real names; names only, no states	Built
Password wall	`proxy.ts`, `lib/server/gate.ts`	One shared password before any page or API route	Built
API and stream	`app/api/`	A stream per viewer, scoped by role; HTTP actions for acks, comments, rules, incidents, imports and exports	Built
Desktop alerts	`lib/server/push.ts`	Web Push to Chrome and Edge through the Google and Microsoft push services	Built; subscriptions in memory
Browser app	`app/(console)/`, `components/`	Console, Dashboards, Instance Ledger, Rules Engine, My View	Built

Part	Where in the code	What it does	Status
PostgreSQL	Proposed	Rules and their versions, call-outs, incidents, notifications, people and teams, shift state, audit, raw events, Genesys regions, orgs, queues and queue membership	Proposed: 20 tables in the PostgreSQL schema doc
SSO	Not built	Who the user is and what role they hold	Planned; the "View as" selector stands in

- **One process, by design.** The engine's state lives in memory, so the app cannot yet run as several copies behind a load balancer. With PostgreSQL it can resume a shift after a restart; scaling out needs one engine process and a broker in front of the web servers.
- **Where the database writes go:** `fire()` and `openIncident()` in `lib/engine/escalation.ts`, and `ack`, `ackAll`, `comment`, `invAction` and the rule setters in `lib/engine/engine.ts`.

Data sources

Genesys Cloud is the only live source today, reached two ways: its REST API and its notification WebSocket. Everything else is an upload, the simulator, the console's own users, or still planned.

Source	How it connects	What it gives the app	Status
Genesys Cloud REST API	HTTPS with a supervisor bearer token (`GENESYS_TOKEN`, or one pasted on `/admin/token`)	Watched queues and their names, queue members with account state and last login, every agent's current presence and routing, queue metrics every 5 s, each agent's handle time every 60 s	Live

Source	How it connects	What it gives the app	Status
Genesys notifications (WebSocket)	A channel opened over REST, then a WebSocket; at most 1,000 topics per channel: two per agent on the roster, plus one per watched queue	Each agent's presence and routing changes as they happen; call end, transfer, hold and after-call work from the queue conversation topic; heartbeats	Live
Floor data template (.xlsx)	Uploaded by an Admin or Manager on Dashboards	Roster with team lead, manager and LOB; a day of statuses; holds; queue intervals	Built; replay only
Genesys agent-status CSV	Uploaded in the same dialog	Statuses only	Built; replay only
Simulator	Runs inside the server	Invented states and queue numbers, on real names from the saved roster	Built; Admin switches it on
Console users	Browser to the API	Acknowledgements, comments, rule edits, incident dispositions, push subscriptions, the "View as" role	Built
NICE IEX (WFM)	Not connected	Adherence and schedules	Planned (data-architecture decision 4)
SSO identity provider	Not connected	The signed-in person and their role	Planned
Org structure source	Not chosen	Team, team lead, manager and LOB for live agents	Undecided (decision 3)
UiPath	Not connected	Batch loads where no API is available, per the RTM Data Tables file	Proposed

- **The token is short-lived.** It is copied from a supervisor's browser and lapses once idle, revoked or signed out (see Token lifetime); a fresh one is pasted on /admin/token without a restart. `GENCLOUD_CLIENT_ID` and `GENCLOUD_CLIENT_SECRET` are read but not used: an OAuth client-credentials grant is not built yet.

- **The watched view decides the floor.** `RTM_VIEW_CONFIG_ID` names a saved Genesys view ("CSBDProviderData", 90 voice queues); its queues and their members are the agents the console watches.

Data points

From Genesys the app gets three kinds of data: who the agents are and what state each is in, what happens on their calls, and five floor-level queue numbers. The console works everything else out itself, and its users add the rest.

Agents, from Genesys

Data point	Genesys source	How often	Used for
Watched queue IDs	`GET /api/v2/analytics/reporting/settings/viewconfigurations/{id}`	At connect	Which queues and agents to watch (90 queues)
Queue names	`GET /api/v2/routing/queues/divisionviews`	At connect	Team names: queues stand in for teams
Agent user ID, name, account state and last login, per queue	`GET /api/v2/routing/queues/{id}/members`	At connect	The roster: 2,743 member accounts; 1,629 after leaving out deactivated accounts and those with no login in 30 days (9 Oct 2026)
Presence: On Queue, Available, Busy, Break, Meal, Meeting, Training, Away, Idle, Offline	`GET /api/v2/users/presences/purecloud/bulk?id=…` at connect (50 a request), then topic `v2.users.{id}.presence`	At connect, then on every change	Agent state, state timers and the duration rules
Routing status: IDLE, INTERACTING, COMMUNICATING, NOT_RESPONDING, OFF_QUEUE	GET /api/v2/users?id=…&expand=routingStatus at connect, only for agents not Offline, then topic `v2.users.{id}.routingStatus`	At connect, then on every change	Agent state

Data point	Genesys source	How often	Used for
Call events: call end, transfer, hold start and end, after-call work	Topic `v2.routing.queues.{id}.conversations`, one per watched queue	On every change	ACW state, Long hold, Short call, Transfer rate, calls handled
Handle time per agent since Eastern midnight	`POST /api/v2/analytics/conversations/aggregates/query` grouped by `userId`, `tHandle`	Every 60 s	Each agent's AHT
Heartbeat	`channel.metadata` messages on the WebSocket, and each successful queue poll	Every few seconds	Feed health and the "Gencloud not responding" rule

Presence and routing combine into one of the console's states: Available, On Call, ACW, Outbound, Aux Break, Aux Personal or Offline. ACW comes from the call events, because routing stays INTERACTING through after-call work (mapping in `lib/feed/gencloud/state.ts`).

Genesys presence and routing (first match wins, top down)	Console state
Presence Offline	Offline
After-call work pending, no call connected	ACW
Routing INTERACTING (an ACD call)	On Call
Routing COMMUNICATING (a non-ACD call, such as a callback)	Outbound
Routing IDLE, or presence On Queue	Available
Presence Break or Meal	Aux Break
Routing NOT_RESPONDING, or any other presence: Available off queue, Busy, Away, Meeting, Training, Idle	Aux Personal

Floor queue numbers, from Genesys

Each is summed or weighted by calls across the 90 watched queues, into one set of numbers for the floor.

Data point	Genesys source and metric	How often	Used for
Calls waiting	`POST /api/v2/analytics/queues/observations/query`, `oWaiting`	Every 5 s	Calls in queue tile, Queue backlog rule
Calls offered and abandoned since midnight	`POST /api/v2/analytics/conversations/aggregates/query`, `nOffered`, `tAbandon`	Every 5 s	Abandon tile and rule
Answer time since midnight	Same query, `tAnswered` (sum and count)	Every 5 s	ASA tile
Service level since midnight	Same query, `oServiceLevel` (calls within target out of calls counted)	Every 5 s	Service level tile and rule
Handle time since midnight	Same query, `tHandle`	Every 5 s	Fetched, not used yet
Agents interacting, on queue and off queue	Observation query, `oInteracting`, `oOnQueueUsers`, `oUserRoutingStatuses`	Every 5 s	Fetched, not used yet

Worked out by the console

Data point	From	Used for
Time in current state, time on hold	A timer started at each state change; at connect, Genesys's own start time for the state (from midnight Eastern at most, except calls) and the hold's real start	Every duration rule
Strikes per agent per rule	Each breach	The escalation ladder: nudge, then TL, then Ops

Data point	From	Used for
Call-outs (instances)	Each rule breach	Instance Ledger, trigger feed, dashboards, desktop alerts
Incidents	The third strike	Incident tiles and the investigation register
Calls handled, AHT, short calls, transfers	Live: call end and transfer from the queue conversation topic, AHT from Genesys's handle time. Replays: the export's call-end events	Short call and Transfer rate rules, agent cards and dashboards

From console users

Data point	Where it is entered
Acknowledgement of a call-out, or of all open ones	Console, trigger feed, desktop alert
Agent's reason (up to 140 characters)	Nudge pop-up or its desktop alert
Rule threshold, severity, route, on/off	Rules Engine
Investigation start, disposition, close	Incident tiles
Data source (Genesys or simulation), Genesys token	Context bar, `/admin/token`
Push subscription per browser	"Enable desktop alerts"

Response payloads

The console reads only a few fields from each Genesys response and turns them into one stream message per second for each browser. The shapes below keep only the fields the code reads; values are examples. Parsing lives in `lib/feed/gencloud/client.ts`, `ws.ts` and `conversations.ts`; the stream types in `lib/types.ts`.

From Genesys REST

Response	Fields the console reads	Becomes
`GET /api/v2/analytics/reporting/settings/viewconfigurations/{id}`	`filter.queueIds[]`	The watched queue list

Response	Fields the console reads	Becomes
`GET /api/v2/routing/queues/divisionviews?id=…`	`entities[].id`, `entities[].name`	Team names (one per queue)
`GET /api/v2/routing/queues/{id}/members?expand=dateLastLogin`	`entities[].id`, `.name`, `.user.state`, `.user.dateLastLogin`; `pageCount`	The roster, less `inactive` accounts and those with no login in 30 days
`GET /api/v2/users?id=…&expand=dateLastLogin`	`entities[].id`, `.dateLastLogin`; only active users come back	Account check for members whose state is missing
`GET /api/v2/users/presences/purecloud/bulk?id=…`	`[].userId`, `.presenceDefinition.systemPresence`, `.modifiedDate`	Startup state and time in state
`GET /api/v2/users?id=…&expand=routingStatus`	`entities[].routingStatus.status`, `.startTime`	Startup state and time in state
`POST /api/v2/analytics/queues/observations/query`	`results[].group.queueId`; `data[].metric` with `stats.count` for `oWaiting`, `oInteracting`, `oOnQueueUsers`, `oUserRoutingStatuses` (qualifier `OFF_QUEUE`)	Calls in queue (sum of `oWaiting`)
`POST /api/v2/analytics/conversations/aggregates/query`, grouped by `queueId`	`results[].group.queueId`; `data[0].metrics[]`: `nOffered.count`, `tAnswered.count` and `.sum`, `tAbandon.count`, `tHandle`, `oServiceLevel.numerator` and `.denominator`	Service level, ASA, abandon rate
The same query, grouped by `userId`	`results[].group.userId`; `tHandle.count` (calls handled), `tHandle.sum` (ms)	AHT = sum ÷ count, per agent

Durations come in milliseconds as `sum` and `count`; `avg` is often missing, so the console divides. The floor numbers sum the raw counts across queues: service level = calls within target ÷ calls counted, ASA = answer time ÷ calls answered, abandon = abandoned ÷ offered.

From Genesys notifications

Every WebSocket message is `{ topicName, eventBody }`. Heartbeats arrive as `channel.metadata`. A presence change:

```
{
  "topicName": "v2.users.<userId>.presence",
  "eventBody": { "presenceDefinition": { "systemPresence": "Break" },
"modifiedDate": "2026-10-09T14:02:11Z" }
}
```

A routing change carries `eventBody.routingStatus.status` (`IDLE`, `INTERACTING`, `COMMUNICATING`, `NOT_RESPONDING`, `OFF_QUEUE`). A queue conversation event is the whole conversation, sent again on every change and once per watched queue it touches; the console reads the agent participants' last call:

```
{
  "topicName": "v2.routing.queues.<queueId>.conversations",
  "eventBody": {
    "id": "<conversationId>",
    "participants": [
      {
        "id": "<participantId>", "purpose": "agent", "userId": "<userId>",
        "calls": [{
          "state": "disconnected",
          "connectedTime": "2026-10-09T14:00:05Z", "disconnectedTime": "2026-
10-09T14:06:59Z",
          "disconnectType": "transfer",
          "held": false, "startHoldTime": null,
          "afterCallWork": { "state": "pending", "startTime": "2026-10-
09T14:06:59Z" }
        }]
      }
    ]
  }
}
```

A call goes `alerting` → `connected` → `disconnected` → `terminated`. Connected to disconnected is a call end, and its length is `disconnectedTime` minus `connectedTime`. `disconnectType` `transfer` counts as a transfer; `peer`, `client` and `error` do not. `afterCallWork.state` is `pending` until the wrap-up is in, then `complete`, or `skipped` (mostly on transfers).

To the browser

`GET /api/stream` sends one `init` message, then a `tick` every engine second and right after any action. Everything is already scoped to the viewer's span.

```
{
  "type": "tick",
  "t": 52130, "mode": "live", "staleFor": 0, "replay": null,
  "queue": { "cq": 14, "sl": 42.8, "asa": 460, "ab": 7.7 },
  "rules": [{ "id": "acw", "name": "Extended ACW", "type": "duration",
"cond": "State ACW longer than", "thr": 120, "unit": "s", "sev": "warn",
"route": "nudge", "on": true }],
  "agents": [{
    "id": 7, "name": "Last, First - AG00000", "team": "<queue name>",
"state": "acw", "stTime": 95,
    "aht": 412, "ahtFromFeed": true, "calls": 9, "shortCalls": 0,
"transfers": 1,
    "onHold": false, "holdTime": 0, "adh": 96, "strikes": { "acw": 1 },
"fired": { "acw": false }
  }],
  "incidents": [{ "inc": "INC-2026-0001", "t": 50210, "agent": "…", "team":
"…", "rule": "Overbreak", "ruleId": "ovbrk", "instances": 3, "status":
"Open", "disposition": "", "closedT": null }],
  "org": [{ "team": "<queue name>", "tl": "", "mgr": "", "lob": "<queue
name>" }],
  "ledger": [{
    "n": 41, "t": 52130, "agent": "…", "isFloor": false, "team": "…", "rule":
"Extended ACW", "ruleId": "acw",
    "val": "2:01", "stage": "nudge", "sev": "warn", "status": "open", "ackT":
null, "strikes": 1,
    "inc": "", "cmt": null, "cond": "State ACW longer than 120s", "rev": 88
  }],
  "toasts": [], "nudges": []
}
```

- `t`, `ackT`, `closedT` are seconds since Eastern midnight; `stTime`, `holdTime`, `aht` are seconds.
- `sl` is `null` until a queue has counted a call: no data, not 0%.
- `ledger` holds the whole scoped ledger in `init` and only new or changed rows (by `n`) in a `tick`.
- `init` adds `view` (role and person), `people` (the "View as" directory, dev flag only), `feed` (`gencloud`, `sim` or `csv`), `realNames` and `canSwitchFeed`.
- A nudge is `{ n, agent, team, first, body }` and goes only to that agent; a toast is `{ kind, title, body, n?, team? }` and goes to leaders with the call-out in span.
- `tl`, `mgr` and `lob` are empty on the live feed: no org structure source yet.

Real or not

On the live Genesys feed, who the agents are, what state each is in and for how long, their calls, holds, after-call work and AHT, and the floor's queue numbers are real; the org structure, adherence, shift schedules, and call counts, strikes and call-outs from before the last restart are not. With **Data** set to **Simulation**, only the names are real: every state, call-out and incident is invented, and the feed pill says so.

Real and live

Shown	From	Refreshed	Checked
Agent names and queue membership	Genesys queue members, 90 watched queues	At connect	1,629 active agents of 2,743 member accounts (9 Oct 2026)
Agent state: Available, On Call, ACW, Outbound, Aux Break, Aux Personal, Offline	Genesys presence and routing: a snapshot at connect, then each change	Within seconds of a change	Counts matched Genesys within a few agents on 8 Oct 2026
Time in current state, including after a restart	The connect snapshot's routing startTime or presence modifiedDate, then the console's own timer	At connect, then every second	1,693 of 1,724 agents seeded on 8 Oct 2026: median 9 min on a call, 13 min on break
Call end and transfer, call length	Queue conversation topic: the agent's call going connected to disconnected, with disconnectType	Within seconds of a change	26 call ends, 3 transfers in 3 min on 8 Oct 2026; median call 416 s
Hold start and end	held and startHoldTime on the agent's call	Within seconds of a change	58 holds started, 29 ended in 3 min
After-call work (ACW state)	afterCallWork on the agent's call: pending, then complete or skipped	Within seconds of a change	33 calls entered ACW in 2 min; most finish in 15 to 120 s

Shown	From	Refreshed	Checked
AHT per agent	Genesys `tHandle` per agent since Eastern midnight (talk, hold and after-call work), on the watched queues	Every 60 s	646 agents with handled calls on 9 Oct 2026
Calls in queue	`oWaiting`, summed across the watched queues	Every 5 s	Matches the queue observations
Service level since midnight	`oServiceLevel`, weighted by calls	Every 5 s	42.8% (4,974 of 11,624 calls) matched a direct query
ASA since midnight	`tAnswered`, weighted by calls	Every 5 s	About 460 s, matched a direct query
Abandon rate since midnight	`nOffered` and `tAbandon`	Every 5 s	7.7 to 7.8%
Feed health	Heartbeats and successful polls	Every second	Drives "Gencloud not responding"
Call-outs from state and queue rules	The engine, from the real states and numbers above	As they happen	Extended ACW, Unscheduled Aux, Overbreak, Long call, Long hold, Outbound call, Short call, the three queue rules and Gencloud not responding

Shown, but not real

Shown	What is really behind it	Effect
Teams	Genesys queues (names like `CSBD_ELV_PROV_CENTRAL_NONPAR_BENEFITS_BOT_AGENT`), with no team lead, manager or LOB	Team Lead and Manager views have no one to scope to

Shown	What is really behind it	Effect
Calls handled, transfers, short calls; AHT before an agent's first handled call	Real calls, but only those seen since the last restart; until Genesys reports a handled call, AHT is the console's estimate	Counts and Transfer rate start at the last restart
Adherence	A default 96% for every agent	Adherence breach never fires
Prolonged offline call-outs	Real offline time, but no shift schedule: logging off at the end of a shift looks the same as going missing mid-shift	A critical call-out to the TL 10 min after anyone logs off
Ledger, strikes, incidents and acknowledgements	Held in memory only	A restart wipes them; history starts at the last restart
Meeting and Training as Aux Personal	Real Genesys statuses, but a mapping choice pending WFM (decision 5)	Scheduled meetings and training can raise Unscheduled Aux

Not yet received

Two of the 14 rules lack an input on the live feed: Adherence breach stays quiet, and Prolonged offline cannot tell off-shift from absent. Five inputs are still to come.

Data point	Needed by	Shown today	Where it would come from
Adherence	Adherence breach	96% for everyone (default)	NICE IEX

Data point	Needed by	Shown today	Where it would come from
Shift schedules	Prolonged offline (mid-shift only); telling scheduled from unscheduled Aux and breaks	Nothing: every offline agent counts as mid-shift	NICE IEX (decision 4)
Team, team lead, manager, LOB	Role scoping, Team Lead and Manager views	Queues as teams, no leads	The org structure source (decision 3)
A credential that does not expire	Running unattended	A token pasted every 8 hours	OAuth client credentials (`GENCLOUD_CLIENT_ID`, `GENCLOUD_CLIENT_SECRET`)
The other orgs and regions	A floor wider than anthem-commercial	One org, 90 queues	The org list per region, each org's watched view and a token for each

1,039 of the 2,743 queue members are deactivated accounts (Genesys state `inactive`), so they stay Offline. None are outside the token's reach. The roster leaves them out, and since 9 Oct 2026 also the 116 active accounts with no login in 30 days; the full list is in `.rtm/inactive-queue-members.csv`, which stays out of the repo.

Engine rules

The engine checks 14 rules every second, on the server. Admin and Manager can change a rule's threshold, severity, route and on/off on Rules Engine; changes apply at once but live in memory, so a restart brings back these defaults (`defaultRules()` in `lib/engine/rules.ts`).

Rule	Type	Fires when (default)	Severity	Route	On live data
Extended ACW	Duration	ACW longer than 120 s	Warn	Nudge → TL → Ops	Works
Unscheduled Aux	Duration	Aux Personal longer than 300 s	Warn	Nudge → TL → Ops	Works; Meeting and Training count as Aux Personal
Overbreak	Duration	Aux Break longer than 900 s	Warn	Nudge → TL → Ops	Works; one limit for every break, no schedule
Prolonged offline	Duration	Offline longer than 600 s	Critical	TL first → Ops	Fires at every logoff: no schedule

Rule	Type	Fires when (default)	Severity	Route	On live data
Long call	Duration	On Call longer than 900 s	Critical	TL first → Ops	Works
Long hold	Duration	On hold longer than 120 s	Warn	Nudge → TL → Ops	Works
Outbound call	Duration	Outbound longer than 600 s	Warn	TL first → Ops	Works
Short call	Event	A call ends in under 30 s	Warn	Nudge → TL → Ops	Works, for calls seen since the last restart
Adherence breach	Duration	Adherence below 90%	Warn	Nudge → TL → Ops	Never fires: adherence is a default 96%
Transfer rate	Ratio	Transfers above 25% of calls, at 5 calls or more	Warn	Nudge → TL → Ops	Works, for calls seen since the last restart
Queue backlog	Queue	More than 8 calls waiting	Critical	TL first	Works
Service level	Queue	Service level since midnight below 80%	Critical	TL first	Works
Abandon rate	Queue	Abandon rate since midnight above 5%	Critical	TL first	Works
Gencloud not responding	System	No heartbeat or successful poll for 30 s	Critical	TL first	Works; not checked in a replay

Who hears about it

Each agent call-out adds a strike for that agent on that rule, and the strike count picks the rung.

Route	Strike 1	Strike 2	Strike 3 and later
Agent nudge → TL → Ops	Nudge to the agent	Call-out to the TL	Ops escalation and an incident
TL first → Ops	Call-out to the TL	Call-out to the TL	Ops escalation and an incident
Agent nudge only	Nudge to the agent	Nudge to the agent	Nudge to the agent
TL only	Call-out to the TL	Call-out to the TL	Call-out to the TL

- **Once per episode.** A duration rule fires once while the agent stays in that state over the limit, and re-arms when the state changes or the value drops back under.

- **Once per shift.** Adherence breach and Transfer rate never re-arm.
- **Offline agents** are checked only by Prolonged offline.
- **Strikes** count per agent, per rule, from the last restart; nothing resets them at shift change yet.
- **Incidents.** The Ops rung opens `INC-2026-NNNN`, at most one open per agent per rule; later strikes raise its instance count. A leader starts the investigation, records one of four dispositions (Coached & documented, Validated / excused, Referred to HR / PIP, System issue / no fault), then closes it.
- **Queue and system rules** carry no strikes and stay on their route's first rung. They fire once per breach, re-arm on recovery, and show as "Queue" on team "Floor". Every leader sees them; agents never do.
- **No flood after a restart.** At connect, rules an agent is already past are marked as fired without a call-out, so only new breaches raise one.
- **Notices.** A nudge goes only to the agent, in the page and as a desktop alert. A TL or Ops call-out goes to leaders with that agent in span. Replays raise no desktop alerts.

Getting the call data

Call end and transfer, hold start and end, and after-call work all come from one Genesys notification topic, `v2.routing.queues.{queueId}.conversations`, on the WebSocket the console already uses. A per-agent aggregates poll, once a minute, adds real AHT. Checked against the tenant's topic list and the current token's permissions on 8 Oct 2026; events were observed live on 8 Oct 2026, and the console now reads them. During ACW, Genesys keeps routing at INTERACTING, so ACW is only visible on this topic.

Recommended topic: `v2.routing.queues.{queueId}.conversations`

One topic per watched queue: 90 topics, against about 1,630 for the per-agent topic. The console holds about 3,260 topics for presence and routing (two per agent on the roster, 4 channels), so this adds one channel.

Data point	Field on the agent's participant	How the console would use it
Call end	`state` becomes `disconnected`, with `disconnectedTime`	Ends the call: its length feeds Short call and AHT
Transfer	`disconnectType: "transfer"`, also `transferType`	Counts toward Transfer rate
Hold start and end	`held: true` with `startHoldTime`, then `held: false`	Starts and stops the Long hold timer

Data point	Field on the agent's participant	How the console would use it
After-call work	`afterCallWork` on the call (state pending, then complete or skipped; start and end time), `afterCallWorkRequired`, `wrapup`	ACW state for Extended ACW; the real end of handle time

Permission: `analytics:conversationDetail:view`. The current token holds it in 7 divisions; whether every watched queue sits in one of them is not yet confirmed.

Alternatives

Endpoint	All four fields	Why it ranks lower
`v2.users.{id}.conversations` (topic)	Yes	2,743 more topics, about three more channels. The fallback if the queue topic misses calls outside the watched queues, such as outbound calls or transfers into unwatched queues
`v2.users.{id}.conversations.calls`, `v2.routing.queues.{id}.conversations.calls` (topics)	No: no after-call work	Lighter, but Extended ACW could not work
`v2.detail.events.conversation.{id}.acw`, `.user.end`, `.wrapup` (topics)	Yes	Keyed by conversation, so each call needs its own subscription
`POST /api/v2/analytics/conversations/details/query`	Yes	Minutes behind: for audits, replays and filling the database, not live rules

Per-agent AHT and transfer rate

`POST /api/v2/analytics/conversations/aggregates/query`, grouped by `userId`, with `tHandle`, `tTalk`, `tHeld`, `tAcw` and `nTransferred`. It returns each agent's real AHT and transfer rate since midnight without counting events; polling once a minute is enough for those two rules. It is the query the console already runs for queues, so the permission

(`analytics:conversationAggregate:view`) is in place. The console now polls tHandle this way once a minute on the watched queues for AHT (about 330 KB at midday; 646 agents with handled calls on 9 Oct 2026). The whole-org version is 1.7 MB, too large to poll. Transfer rate still counts the transfers the console sees.

Before building

- [x] ~~Subscribe to the queue topic for a few minutes while the floor is busy, and confirm what a transfer, a hold and after-call work look like in real events.~~
- [ ] Confirm the watched queues sit in the token's 7 divisions.
- [ ] Confirm the tenant's channel limit per user: the console holds 5 channels.

Getting time already in state

No new permission is needed: the startup snapshot carries when each agent's presence and routing status began, and the console seeds each agent's timer from them. Since 9 Oct 2026 it reads presence in bulk, `GET /api/v2/users/presences/purecloud/bulk?id=…` (50 a request, about a tenth of the data), and routing status, `GET /api/v2/users?id=…&expand=routingStatus` , only for agents not Offline. Genesys has no bulk routing-status read.

Field	Meaning	Present
`presence.modifiedDate`	When the current presence (Break, Busy, Offline…) was set	100 of 100
`routingStatus.startTime`	When the current routing status (IDLE, INTERACTING…) began	78 of 100; missing mostly for users who have not routed recently

Time in state at connect is now minus the timestamp behind the state shown:

- **On Call, Outbound, Available** come from routing: use `routingStatus.startTime` .
- **Aux Break, Aux Personal, Offline** come from presence: use `presence.modifiedDate` .
- Where `startTime` is missing, fall back to `modifiedDate` .

The live updates carry the same fields (`modifiedDate` on presence messages, `routingStatus.startTime` on routing messages). Today the console starts each timer when a message arrives; using Genesys's timestamp also removes the delivery delay.

Other sources

Endpoint	What it gives	Use it for
`POST /api/v2/analytics/users/details/query`	Each agent's full status history for the day, with start and end times	Backfilling a day's timeline, or checks against adherence
`POST /api/v2/analytics/users/aggregates/query`	Total time per presence and routing status in an interval	Daily totals, not the length of the current stretch
`GET /api/v2/users/{id}/presences/purecloud`, `GET /api/v2/users/{id}/routingstatus`	The same two fields, one user at a time	Spot checks

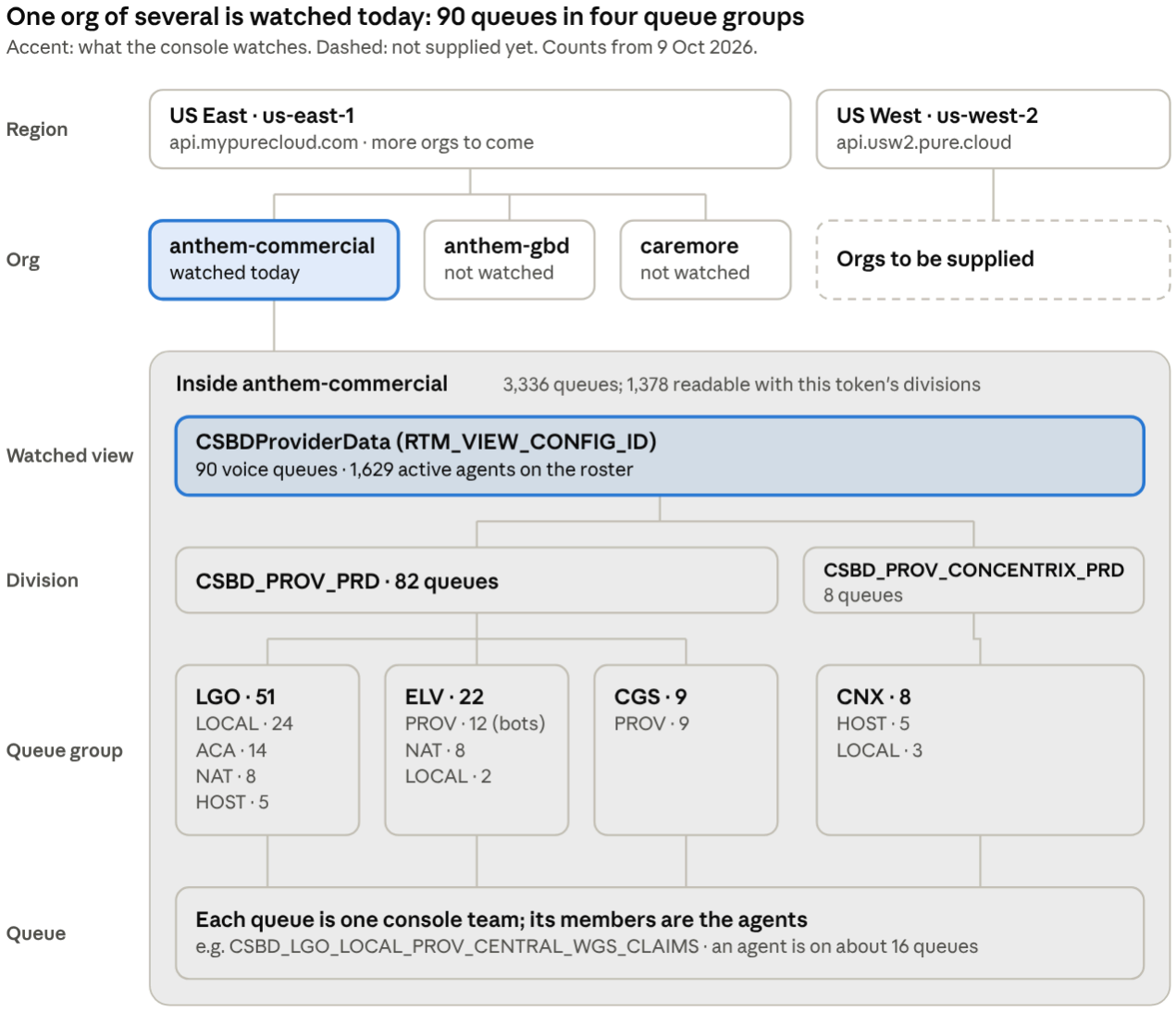
Before switching it on

With real timers, a restart would fire at once every rule an agent is already past. In the sample, the median agent had been in their state for 589 minutes, mostly Offline overnight, so Prolonged offline would raise a critical call-out for nearly every offline agent, and Unscheduled Aux and Overbreak for anyone already over.

- [x] ~~On restart, mark rules already in breach as fired without raising call-outs. The engine already does this for the simulator's warm-up (`rearm()` in `lib/engine/engine.ts`).~~
- [ ] Get shift schedules (NICE IEX, decision 4) before Prolonged offline mid-shift can tell off-shift from absent; until then that rule cannot work on live data either way.

Regions, organizations, queue groups and queues

The organization is the unit the console connects to: each org has its own token, users, queues and rate limit, and its region only decides which hosts the calls go to. There are N regions, each with several orgs. Inside an org, divisions decide what a token can read, a saved view picks the watched queues, and the queue names fall into families, here called queue groups. Today the console watches one org, `anthem-commercial` in US East.



regions, orgs, watched view, divisions, queue groups, queues

Only anthem-commercial is connected; inside it, the saved view picks 90 of the 1,378 queues the token can read, across two divisions and four queue groups.

Region	API host	Login host	Orgs
US East (us-east-1)	`api.mypurecloud.com`	`login.mypurecloud.com`	`anthem-commercial`; anthem-gbd; caremore; the rest to be supplied
US West (us-west-2)	`api.usw2.pure.cloud`	`login.usw2.pure.cloud`	To be supplied

What changes in the console

- **One feed per org.** Each `GencloudFeed` has its own token, request budget, notification channel and health. A merging feed combines the rosters and queue numbers.
- **Agents keyed by org and user ID.** One person belongs to one org (to be confirmed), so the person record carries its org.
- **Health per org, grouped by region.** "Gencloud not responding" fires per org; the pill reads, for example, "US West · 1 of 3 orgs not responding".
- **Floor queue numbers combine from raw counts:** calls within target over calls counted, and total answer time over calls answered.
- **"Today" starts at Eastern midnight** (America/New_York) for every org. Built on 8 Oct 2026 (RTM_TIME_ZONE); before, it was the server's Manila midnight.
- **Tokens per org.** `/admin/token` gains an org picker until OAuth client credentials replace pasted tokens.

Token lifetime

The pasted token has no fixed expiry: Genesys reports when it goes idle instead. `/api/v2/tokens/me` gives `dateTokenIdles` 2026-10-14 08:06 UTC for the current token, which is why the one from a few days ago still works. It still lapses if left unused past that date, revoked, or signed out, so OAuth client credentials remain the fix for unattended running.

Queues and people in anthem-commercial

Counted on 9 Oct 2026 with the console's token, read-only.

Scope	Queues	Memberships	People
Whole org (names and IDs only)	3,336		
Readable with this token (its divisions)	1,378 (1,329 with members, 49 empty)	122,296	
Watched view (`RTM_VIEW_CONFIG_ID`)	90 (7 to 1,429 members each, median 584)	44,386	2,743 accounts: 1,039 deactivated, 116 with no login in 30 days, 1,629 on the roster at the time of the count (9 Oct 2026)

Scope	Queues	Memberships	People
Users in the org			10,274 active; 26,090 in any state

- A membership is one person on one queue. Agents are on many queues, so memberships far outnumber people: the watched view's 44,386 memberships are 2,743 accounts, about 16 queues each.
- The token reads members and settings only for queues in divisions its user can see; the other 1,958 are visible by name only. Watching any of those needs a token with wider division access.
- The view held 89 queues when the feed was built and 90 now: the queue list is edited in Genesys, and the console picks the change up at its next restart.

Queue groups in the watched view

A queue group is the family a queue's name puts it in: the second and third parts of `CSBD_<group>_<subgroup>_…`. It is not a Genesys object and not yet a console level: the console shows each queue as a team and sums all 90 into one set of floor numbers. Counted from `.rtm/genesys-queues.csv` (written by `tools/list-queues.mjs`) on 9 Oct 2026.

Group	Subgroups (queues)	Division	Queues
LGO	LOCAL (24), ACA (14), NAT (8), HOST (5)	CSBD_PROV_PRD	51
ELV	PROV (12, all `_BOT_AGENT`), NAT (8), LOCAL (2)	CSBD_PROV_PRD	22
CGS	PROV (9)	CSBD_PROV_PRD	9
CNX	HOST (5), LOCAL (3)	CSBD_PROV_CONCENTRIX_PRD	8

The rest of a name narrows it further: region (CENTRAL, EAST, NY), platform (WGS, FACETS, CS90), PAR or NONPAR, BENEFITS or CLAIMS. An agent sits on about 16 queues, often across groups, so a person cannot be placed in one group by queue alone.

Decisions

Decision	Status
Orgs in each region, with each org's watched queues or view	Adrian to supply
One person per org	Assumed; Adrian to confirm
Time zone for "today"	Decided: Eastern; built 8 Oct 2026
One OAuth client per org	Open
One merged floor, or filters by region and org	Open
Queue numbers and filters per queue group, or one set for all watched queues	Open (data-architecture decision 6)